1. Trained classification models

Your notebook trains five machine learning models:

- Random Forest
- K-Nearest Neighbors (KNN)
- Decision Tree
- Naive Bayes
- Support Vector Machine (SVM)

The tuned models are available in your notebook's best_models dictionary. They are trained in the notebook, but they are not currently saved as separate model files.

2. Evaluation metrics and comparison results

These are the actual results from your notebook after hyperparameter tuning.

Classification model	Test accuracy	Precision	Recall	F1-score
Random Forest	96.35%	96.45%	96.35%	96.34%
KNN	51.14%	51.14%	51.14%	51.14%
Decision Tree	99.25%	99.25%	99.25%	99.25%
Naive Bayes	67.26%	69.72%	67.26%	66.23%
SVM	99.89%	99.89%	99.89%	99.89%

3. Final conclusion identifying the best model

Conclusion

In this project, five classification algorithms were implemented and evaluated to predict crowdfunding campaign success. The models were compared using accuracy, precision, recall, F1-score, cross-validation, and hyperparameter tuning.

Among the evaluated models, Support Vector Machine (SVM) achieved the best performance, with a test accuracy of 99.89% and a weighted F1-score of 99.89%. The Decision Tree classifier was the second-best model, achieving a test accuracy of 99.25%.

Therefore, SVM was identified as the best-performing model for this experiment. However, further validation is required before applying it to real-world predictions, as features such as the amount raised may reveal information about the campaign's final outcome.

This project strengthened my practical understanding of data preprocessing, classification algorithms, model evaluation, cross-validation, and hyperparameter optimization as part of my journey as an aspiring Data Scientist.